



Today, many students who do not get quality education in the colleges are depending on vocational training imparted by training institutes. Such trainings can help in strengthening a student's resume from a job-search perspective. But training institutions cannot lay the foundation, they can only build

upwards — Dr C.P. Ravikumar, technical director, University Relations, Texas Instruments India

programmes and so on," explains Dr Ravikumar.

Vivek Sharma, regional vice president, GCSA-India operations and director, India Design Centres, STMicroelectronics, adds, "In India, there are some institutes which are world-class and then there are institutes which are not good at all. There is huge difference between the engineering graduates from these institutes. Good talent is coming only from a few institutes; many others are producing a good proportion of engineers but there is not enough percentage of good talent."

Sharma continues, "It is very important that the students understand the ecosystem of the market. For this, universities and institutes need to work closely to understand the needs of the market in coming times. The universities should focus on research activities that the industry would need. To address this issue, industries can initiate bringing the university trainees and have them work for some time in their environment. Also, they can jointly set up research labs so that industries can work with the academia on projects for the market. I strongly believe it as a very meaningful way to bring the need of the industry to the university."

Indrajit Sabharwal, managing director, Simmtronics Semiconductors, suggests for engineering colleges that "Fifty per cent of the course which is imparted should be knowledge-based, 25 per cent should be industry-applicable and the rest must be industry knowledge and training." "Some engineering institutes are following

this practice and are successful too," he adds.

Use of technology can improve the situation. Sharma says, "Institutes usually have links with very good faculty. With technology, their accessibility can be increased. For example, some institutes which are far off can be connected and one can listen to lectures of the most talented faculty through technology. Technologies can help the institutes which are missing on good faculty and also to the institutes which don't even have the faculty. This is one way with which we can bridge the gap. Also, universities can regularly look into the curriculum to make it more contemporary—not to say that universities should not look into the basic fundamentals. Strong fundamentals are what I strongly advocate and skills can be further nurtured by the industry. Many times universities focus on skills which can be acquired rather than the real fundamentals."

Skills to focus

Keeping in mind the skill requirements of the industry, training institutes offer various types of specialised vocational courses in very large-scale integration (VLSI), microcontrollers and embed-

ded systems, electronics design, automation, surface-mount technology (SMT), telecom, and hardware maintenance and repair. Through short-term courses, they prepare engineers to handle projects independently, thus enabling companies to induct new staff in a project with minimal domain-specific training.

The demand for employable individuals is quite high in the telecom sector, which is spread across operators, integrators and vendors. Professionals with science and engineering background often need specialised training in theory and practice so that they can properly handle different types of machines and instruments in various telecom systems, both wired and wireless. Some of the niche skill sets in demand are advanced mobile technologies like 3G, mobile repairing techniques and networking.

In the automation sector, candidates require industry-focused training on various industrial automation systems like programmable logic controllers (PLCs), SCADA, HMI, drives, PLC networking, process instrumentation and panel designing. Companies in this sector are large with highly technical processes, and expect professionals to have the knowledge of complex industrial automation control systems in order to reduce the downtime of machines and troubleshoot the systems faster, helping them keep the productivity high.

Candidates willing to enter the semiconductor industry should possess technical skills in embedded systems, microcontrollers, VLSI design and sometimes in niche areas like



Realising the need for practical training in specific fields that are important to the industry, we offer hands-on training in certain subjects. Learning is through carefully designed presentations and lab exercise. The attempt is to fill the gap by covering what has been missed in the classroom — Col. (Retd) N.C. Pande, joint director (training), EFY Tech Center